

① UDP is connectionless, so the client & server do not establish a connection before communication.

UDP Server Sequence

```
socket()
↓
bind()
↓
recvfrom() client
↓
sendto() client
↓
close()
```

UDP Client Sequence

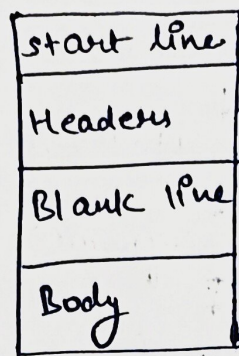
```
socket()
↓
sendto() server
↓
recvfrom() server
↓
close()
```

The UDP client doesn't need bind() because the OS can automatically assign a temporary port.

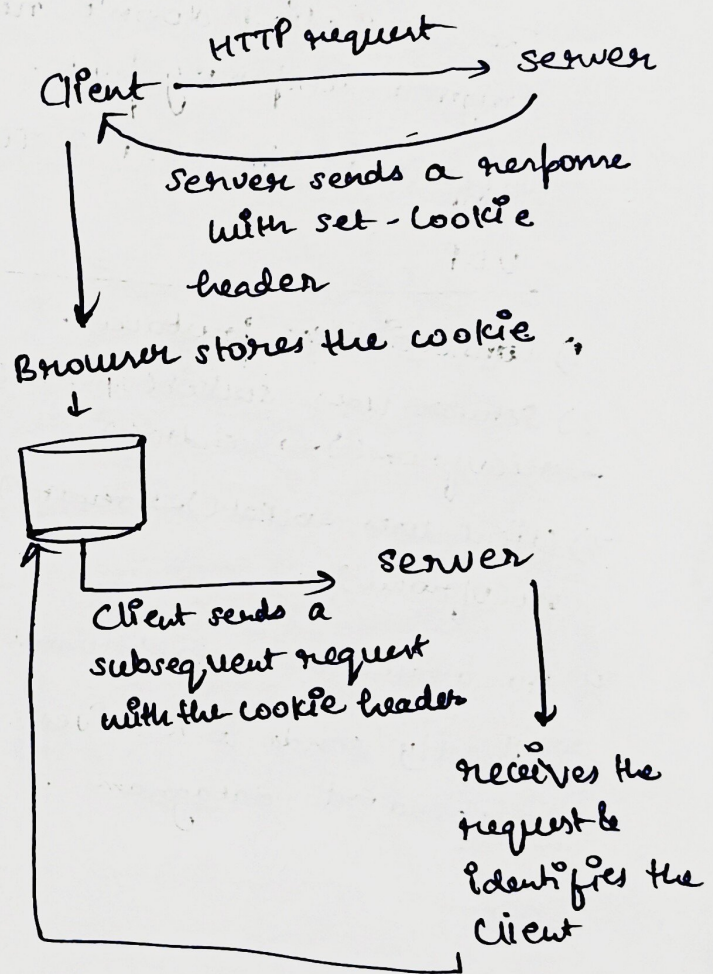
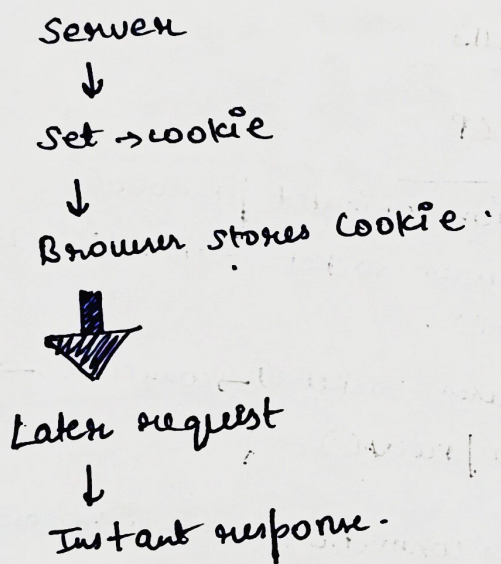
Difference b/w UDP & TCP socket calls

UDP	TCP
1) Connectionless protocol.	1) Connection-oriented protocol.
2) Server uses socket() → bind() → recvfrom() → sendto()	2) Server uses socket() → bind() → listen() → accept().
3) Client uses socket() → sendto() / recvfrom()	3) Client uses socket() → connect() → send() / recv().
4) No connection establishment.	4) Requires connection establishment.
5) Directly sends & receives independent datagrams.	5) Can not directly send / receive independent datagrams.

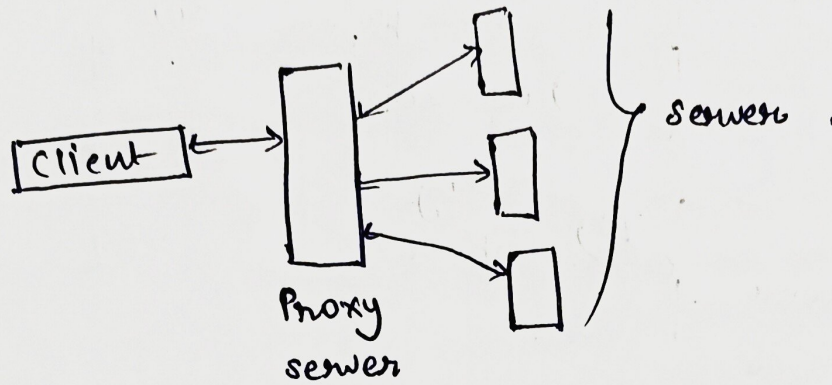
(2) General structure of an HTTP message consists of 4 main parts: start line, headers, a blank line & a body.



(3) Cookies allow a server to store a small amount of information on the client & use that information in later requests.



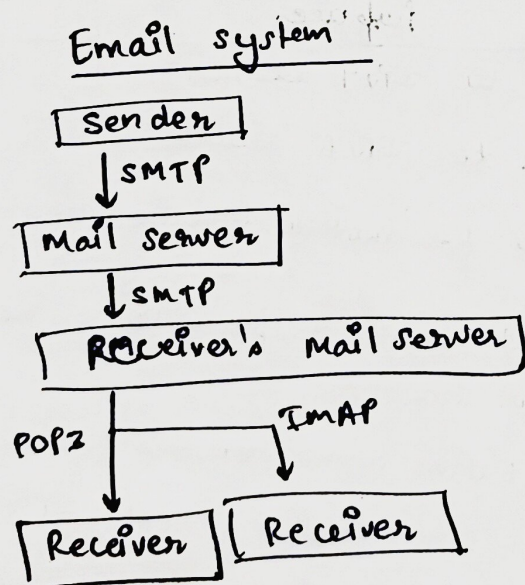
(4) A proxy-server is an intermediate server that acts between a client & the destination server. It receives requests from a client, forwards those requests to the destination server, receives the response & then sends the response back to the client.



(5) Both methods are used for HTTP cache validation. The client asks the server whether its cached copy is still valid.

Last - modified	ETag
(1) Uses the modification date/time	(1) Uses a resource identifier.
(2) Last-modified / If-modified-since	(2) ETag / If-none-match.
(3) Depends on timestamp accuracy	(3) Directly identifies a particular representation.
(4) Can miss changes occurring within the same timestamp.	(4) Can detect changes even when modification time is unchanged.

(6)

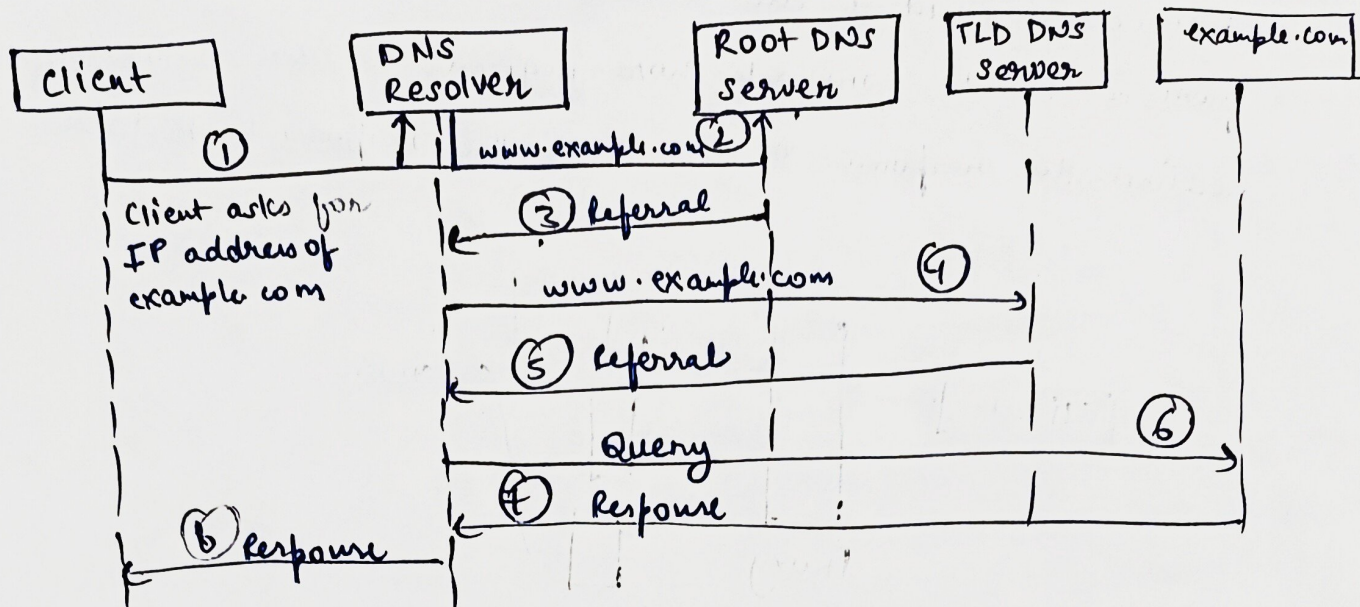


1) SMTP is used to send email. It is mainly a sending / transfer protocol.

2) POP3 is used by a client to retrieve emails from the mail server.

3) IMAP - allows a client to access & manage emails while they remain stored on the mail server.

(7)



(9) Optimistic unchoking is a BitTorrent mechanism where a peer temporarily unchokes a randomly selected interested peer, even if that peer has not yet been uploading much data in return. In the standard algorithm, this slot rotates periodically, commonly every 30 seconds. The normal fit-for-tat mechanism mainly rewards peers that are already uploading data to you. This creates a problem for a new peer. It may have no pieces to upload yet, so established peers have little reason to unchoke it.

(8) Record Type	Purpose
A	Maps a domain to IPv4 address.
AAAA	Maps a domain to IPv6 address.
CNAME	Create an alias for another domain name.
MX	Specifies the mail servers responsible for receiving email for a domain.
NS	Specifies authoritative name servers for a domain.
PTR	Performs reverse DNS lookup, mapping IP to domain.
TXT	Stores text information associated with a domain.
SOA	Start of Authority record, contains important administrative information about a DNS zone.

(10)

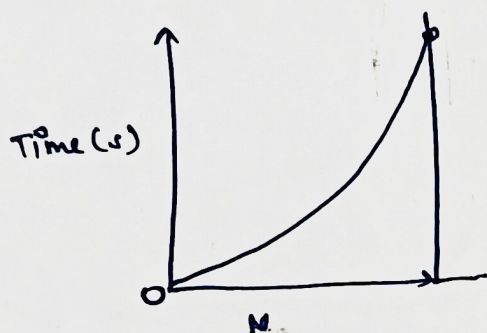
$$\text{File size (F)} = 20 \text{ Mb} = 20 \times 10^6 \text{ bits}$$

$$\text{Server upload rate } (\mu_s) = 400 \text{ kbps} = 400 \times 10^3 \text{ bits/s}$$

$$\text{Client download limit} = \frac{20 \times 10^6}{200 \times 10^3} = 100 \text{ s}$$

for $N = 5$

$$D_{cs} = \frac{5 \times (20 \times 10^6)}{400 \times 10^3} = \frac{10^3}{4} = 250 \text{ s}$$



The distribution time increases linearly with the number of clients because the server must upload one complete copy of the file to every client.

(11) For P2P distribution, the minimum distribution time is :-

$$\max \left[\frac{F}{\mu_s}, \frac{F}{d_{\min}}, \frac{NF}{\mu_s + \sum_{i=1}^N \mu_i} \right]$$

given :- $\mu_i = 100 \text{ kbps}$

$$\sum \mu_i = N \times 100$$

$$\text{File size} = 20 \times 10^6 \text{ bits}$$

$$\mu_s = 400 \text{ kbps}$$

$$d_{\min} = 200 \text{ kbps}$$

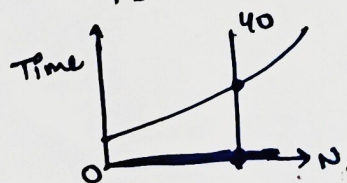
$$\mu_i = 100 \text{ kbps}$$

$$\frac{F}{\mu_s} = \frac{20 \times 10^6}{400 \times 10^3} = 50 \text{ s}$$

$$\frac{F}{d_{\min}} = \frac{20 \times 10^6}{200 \times 10^3} = 100 \text{ s}$$

$$\frac{NF}{\mu_s + N\mu_i} = \frac{20N \times 10^6}{400 \times 10^3 + 100N \times 10^3} = \frac{200N}{N+4}$$

$$D_{P2P} = 100 \text{ seconds}$$



$$\boxed{P2P = 100 \text{ s for all given } N}$$

(12) (1) Difference :-

Feature	Non-Persistent HTTP	Persistent HTTP
Connection	A new TCP connection is created for each object.	A single TCP connection can be used for multiple objects.
RTT Overhead	Higher	Lower
TCP connection for this page.	5 connections : 1 HTML + 4 images	1 connection.

(2) Number of objects = 5
 RTT = 100 ms
 Transmission time = negligible.

→ Non-Persistent HTTP

Time per object = 2RTT

for 5 objects = $5(2RTT) = 5 \times 2 \times 100 = 1 \text{ sec}$

→ Persistent HTTP

for base HTML = 2RTT

$T_{\text{Persistent}} = 2RTT + 4RTT = 6 \times 100 = 0.6 \text{ sec}$

(13) A SMTP email message generally consists of three main parts →
 Header, Blank line & message body.

